

Course Syllabus

Professor: Dr. Christine Heitsch

Office: Skiles 211B

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Office Hours: After class MW from 3:15 - 3:45, and Thursday from 4:30 - 5:30pm online via <https://gatech.zoom.us/my/ph4048944758>.

Individual Appointments: To discuss performance in the class or any other private concerns, send a message through Canvas Inbox to make arrangements.

Piazza Forum: Ask questions about course format and/or content here — i.e. if the answer would be useful to any other student — so that the information is available to all. Postings can be anonymous (to classmates), and should be respectful and courteous at all times.

2106-B Lecture Assistant (LA): Julián Villaquirá. He will have office hours in the Math Lab (Clough 280). For more information, including full schedule, look under “Academic Resources” at <https://math.gatech.edu/undergraduate/current-students>.

Lectures: MW from 2:00 – 3:15pm in Skiles 170.

Textbook: G. Chartrand, A. D. Polimeni, and P. Zhang. *Mathematical Proofs: A Transition to Advanced Mathematics*, Fourth Edition, Pearson, 2018. *Any prior edition should also be fine.*

Course Description: “An introduction to proofs in advanced mathematics, intended as a transition to upper division courses including Abstract Algebra I and Analysis I.”

Prerequisites: Math 1552: Integral Calculus or equivalent.

Course Topics: Fundamentals of mathematical abstraction including sets (Ch. 1), logic (Ch. 2), equivalence relations (Ch. 9), and functions (Ch. 10). Thorough development of the basic proof techniques (Ch. 3 – 8): direct, contrapositive, existence, contradiction, and induction. Introduction to proofs in algebra (Ch. 15) and analysis (Ch. 14). Selected additional topics (from Ch. 0, 11, 12 as well as partial orders, upper/lower bounds, and $\limsup/\liminf$).

Learning Outcomes: Students should be able to understand and prove simple mathematical statements. For instance, in analysis, students will be able to prove that a given sequence diverges to infinity. In algebra, students will be able to prove that up to isomorphism there are two distinct groups of order four.

Grading Scheme: Grades will be based on **exams (80%)** and **assignments (20%)** according to the standard scale: **A** (90 – 100), **B** (80 – 89), **C** (70 – 79), **D** (60 – 69), **F** (0 – 59).

Exams: There will be four hourly exams, each worth 20% of the final grade. Exams will be closed book, closed notes, no calculator, individual tests. No makeup exams are scheduled.

Three exams will be in-class. Exam dates will be reconfirmed at least a week in advance. Hourly 1 is scheduled for Wednesday **Sep 10**, Hourly 2 for **Oct 8**, and Hourly 3 for **Nov 5**. Hourly 4 will be during the final exam period from **2:40 – 3:55pm** on Friday, **Dec 5**.

Assignments: There will be three types of assignments — in-class, online, and homework — worth a total of 20% of the final grade. Assignments will be due regularly, including on “Final Instructional Class Day.” To succeed in this class, you must be able to demonstrate **mastery** of the material; assignments are a chance to **practice** for the exams.

ICAs: In-Class Assignments will be given most days, and worth 5% of the total course grade. They will be graded on a scale of 0 – 2 based on effort only, and the lowest 2 scores will be dropped when computing the average.

WeBWorK: Online problem sets will be assigned through Canvas, and worth 5% of the total. No partial credit will be given, and the lowest score will be dropped when computing the average.

Homework: Written homework will be assigned most weeks, and worth 10% of the total. It will typically be due a week later, and solutions must be turned in via Gradescope in Canvas.

Late homework will be accepted with a penalty of 5% per 6 hours, up to a total of 36.

If 85% of students complete the class CIOS survey, one homework score will be dropped from the grading scheme computation. Otherwise, all scores will be used.

A subset of the assigned problems will be selected for grading. For full credit, all directions must be followed. **Illegible and/or unintelligible solutions will receive no credit.**

Collaboration with classmates is allowed (and actually strongly encouraged!) when working on homework problems and other coursework.

However, solutions must adhere to the following content guidelines.

1. Be written independently in a student’s own words.
2. Clearly acknowledge any person with whom any part of the assignment was discussed.
3. Properly credit **ANY**¹ outside resource consulted in completing the assignment.

Any solution which violates the content guidelines will receive no credit. Flagrant or repeated violations will be dealt with as a matter of academic integrity.

L^AT_EX: Homework solutions which are typeset in L^AT_EX will receive an additional 10% bonus points, up to a maximum of 100% per assignment. Problems will be provided in a *.tex source file. Georgia Tech provides access to Overleaf Professional, an online browser based L^AT_EX editor: <https://www.overleaf.com/edu/gatech>.

Attendance: Regular participation is expected. Exceptions will be accommodated only for valid, documented reasons including (1) official representation of the Institute and (2) medical or personal emergencies (as verified by the Dean of Students).

Exceptions: Any student who may not be able to meet the requirements of the class as stated must contact the professor individually within the first two weeks of class.

Academic Integrity: Students are expected to abide by the GT *Academic Honor Code, Student Code of Conduct*, and other policies available online through the Office of Student Integrity. **Any violations must be reported directly to the Dean of Students.**

Updates: This syllabus is subject to modification. Any changes will be announced in Canvas.

¹GenAI, e.g. ChatGPT, MathGPT, etc., is most definitely an “outside resource”.